Are Yu-Gi-Oh! Cards Sharing the Same Type More Likely to Interact Together?

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Abstract — This study investigates whether pairs of Yu-Gi-Oh! cards sharing the same monster type are more likely to interact together, using a dataset of annotated card pairs produced by a large language model (Llama 3.1 8B). We fitted a multivariate regression ($R^2 = 0.02$) on monster-pairs, modelling the number of interactions between two cards (*interactCount*) as a function of shared type, shared attribute, shared frametype, level difference, attack difference, defence difference, and annotation-quality control features. Shared type ($\beta = 0.18, p < 0.001$) and shared attribute ($\beta = 0.13$, $p < 0.001$) emerge as the strongest positive predictors, while combat-stat differences show no significant effect. Annotation-quality controls — *stability* ($\beta = -0.68, p < 0.001$) and *repeated* ($\beta = -0.34, p < 0.001$) — carry large negative coefficients, indicating that the model's variability introduces noise that should be accounted for when interpreting the main effects.

Keywords — Yu-Gi-Oh!, card interactions, deck synergy, LLM annotation, Llama 3.1 8B, multivariate regression.

I. Introduction

Yu-Gi-Oh! is a Japanese trading card game in which players assemble decks of 40 – 60 cards and compete by summoning monsters, activating spell and trap effects, and reducing the opponent's life points to zero. With over 14,000 unique cards in circulation, understanding which cards work well together is one of the most demanding cognitive tasks faced by competitive players.

Deck synergy — the capacity of a set of cards to reinforce one another through their effects — is typically evaluated through years of gameplay experience. In this study, we take a data-driven approach to determine whether a simple structural feature of a card, namely its **monster type** (Dragon, Warrior, Spellcaster, etc.), is a statistically significant predictor of whether two cards interact.

This question is motivated by the observation that the majority of support cards in the game are formulated around the monster type. A card that reads *"Add 1 Warrior monster from your Deck to your hand"* explicitly targets cards sharing the Warrior type. If this design principle is reflected in the data at scale, we would expect same-type pairs to have significantly more interactions than cross-type pairs.

Question. *Are monster cards sharing the same type significantly more likely to interact?*

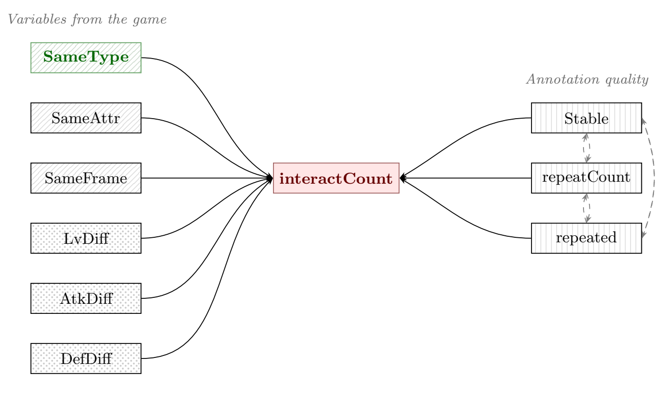


Fig. 1: DAG representing our theory.

II. Data

A. Card Database

Card data was extracted from the YGOProDeck API [4], providing a dataset of 14,137 cards [3]. Each card is described by its name, effect text, frametype (normal, effect, fusion, synchro, xyz, link, spell, trap, etc.), monster type, attribute (DARK, LIGHT, FIRE, WATER, WIND, EARTH), level, attack, and defence. A full description of all variables is provided in Table A1.

B. Interaction Dataset

A second dataset of 27,417 directed card pairs $(A \to B)$ [2] was annotated by an LLM (Llama 3.1 8B) prompted with domain-specific instructions (see Appendix C). For each pair, the model produced a binary label for each of 56 interaction types (e.g. `special_summon`, `destroy`, `target`, `references_type`). A subset of pairs was annotated multiple times because of the probabilistic nature of sampling process (see Appendix B, Fig. A1).

III. Method

A. Feature Engineering

Starting from the raw interaction dataframe, we construct one observation per unique directed pair. The response variable *interactCount* is the mean total number of active interactions across all annotated occurrences of that pair:

$$interactCount_p = \frac{1}{|R_p|} \sum_{r \in R_p} \sum_{j=1}^{56} x_{rj} \qquad (1)$$

where R_p is the set of annotation rows for pair p and $x_{r,j}$ is the binary label for interaction type j in row r.

Two annotation-quality variables are computed. *repeated* is a binary indicator equal to 1 if a pair was annotated more than once. *stability* equals 1 if all repeated annotations agree, 0 otherwise.

Six card-level features are derived from card data:

- **SameType** — 1 if cards A and B share the same monster type, else 0.

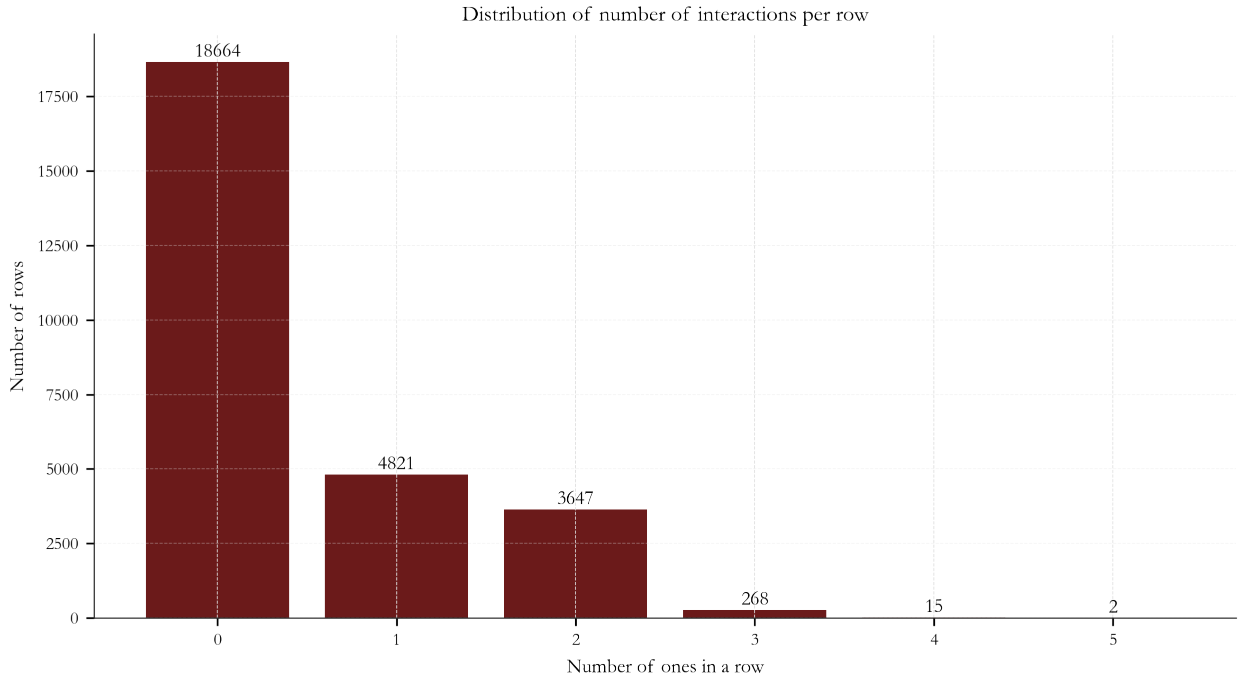


Fig. 2: Distribution of the number of active interactions per annotated row. The distribution is right-skewed: most pairs share zero or one interaction, while a small subset shares several.

- **SameAttr** — 1 if cards A and B share the same attribute, else 0.
- **SameFrame** — 1 if cards A and B share the same frametype, else 0.
- **LvDiff**, **AtkDiff**, **DefDiff** — absolute differences in level, ATK, and DEF.

Pairs involving at least one spell or trap card are excluded, retaining 9,518 monster pairs. Continuous predictors are standardised (mean 0, standard deviation 1). The causal structure assumed by our model is represented by the DAG in Fig. 1.

B. OLS Model

We estimate the mean number of interactions per pair as a linear function of the nine predictors described above:

$$\textit{interactCount}_i = \beta_0 + \sum_{k=1}^{9} \beta_k \, x_{k,i} + \varepsilon_i \tag{2}$$

where $x_{1..9}$ are, in order, *SameType*, *SameAttr*, *SameFrame*, *LvDiff*, *AtkDiff*, *DefDiff*, *stability*, *repeatCount*, and *repeated*. The model is estimated using `statsmodels` (Python 3). Coefficients, confidence intervals, and p-values are reported and visualised as a forest plot.

IV. Results

A. Descriptive Statistics

Fig. 2 shows the distribution of *interactCount* across all annotated rows: right-skewed, with the majority of pairs sharing zero or one interaction. Fig. A2 shows the distribution of binary predictors in the monster-pair subset: same-type pairs are the least frequent (13.0%), while same-attribute (24.8%) and same-frametype (52.2%) pairs are more common.

B. Regression Results

Table 1 summarises the estimated coefficients ($n = 9{,}518$, $R^2 = 0.024$). The forest plot in Fig. 3 visualises the confidence intervals for each predictor.

Table 1: OLS estimates. Significant predictors ($p < 0.05$) are bolded.

Predictor	Coef.	p-value
Intercept	**1.179**	**< 0.001**
SameType	**0.178**	**< 0.001**
SameAttr	**0.133**	**< 0.001**
SameFrame	**−0.036**	**0.026**
LvDiff	0.008	0.413
AtkDiff	0.000	0.974
DefDiff	0.008	0.366
stability	**−0.682**	**< 0.001**
repeatCount	**−0.023**	**0.027**
repeated	**−0.337**	**< 0.001**

C. Interpretation

SameType is the strongest positive predictor: same-type pairs have on average 0.18 more interactions than cross-type pairs, after controlling for all other variables. **SameAttr** shows a comparable positive effect. **SameFrame** carries a small but significant *negative* coefficient, reflecting the fact that pairs drawn from the same frametype (e.g. two normal monsters) often interact through external support cards rather than directly. Combat-stat differences (*LvDiff*, *AtkDiff*, *DefDiff*) are all non-significant. The large negative coefficients on *stability* and *repeated* indicate that repeated annotations tend to converge towards lower interaction counts, reflecting LLM variability in borderline cases; this motivates including these controls rather than ignoring them.

V. Discussion & Limitations

Our results are consistent with the design philosophy of Yu-Gi-Oh!. The game's support cards are systematically formulated around monster types, which naturally creates type-homogeneous interaction clusters.

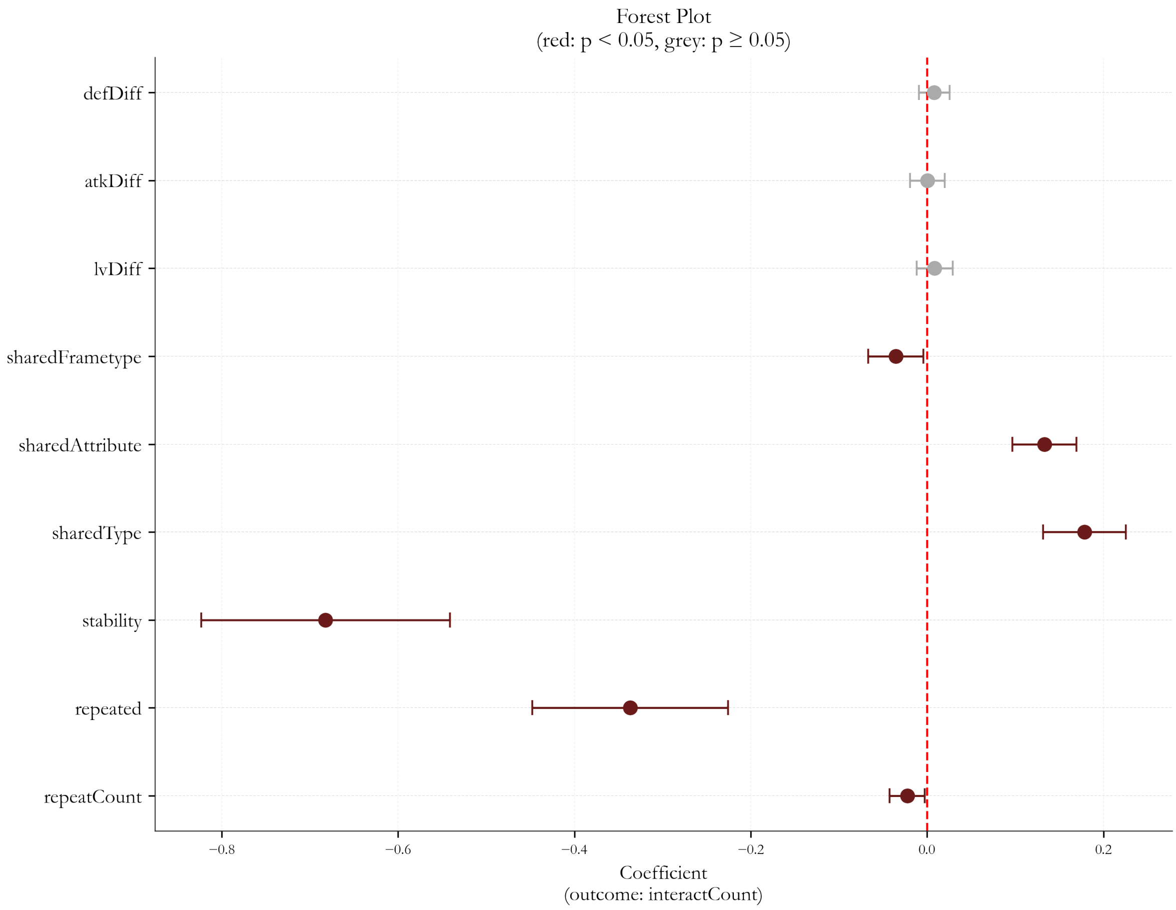


Fig. 3: Forest plot of OLS coefficients with confidence intervals. Red points: $p < 0.05$. Grey points: non-significant. Red dashed line: zero effect.

Several limitations apply. The dataset covers only a fraction of the complete pairwise space (27,417 annotations out of more than 10^8 possible pairs), and its representativeness cannot be guaranteed. The assumption that LLM annotations approximate human-quality labels cannot be formally verified without gold-standard human annotations (that I am working on, through a collaborative annotation website project); the significance of the annotation-quality controls indicates real noise in the labelling process. Finally, the model explains only a modest fraction of variance in *interactCount* ($R^2 = 0.02$), indicating that card-effect text — not captured in our feature set — accounts for most of the remaining variation.

VI. Conclusion

This study demonstrates that **shared monster type is a significant positive predictor of card interaction count** in Yu-Gi-Oh!, after controlling for shared attribute, shared frametype, combat-stat differences, and LLM annotation quality. The result is directly interpretable for a player: building a deck around a single monster type maximises the probability that your cards interact with one another.

Future work could incorporate archetype (that is very difficult to define here), a rule-based relevant pairs selector (such as "in conditions" used in this last study [1]) to extract more cases of interactions and start use interaction types, or use graph-based methods to identify the most central cards within type-specific interaction networks. Graph-based analytics seems to be very promising in understanding underlying structures of the game, and quantify deck synergy.

References

[1] Farès Blibeche. Are Yu-Gi-Oh! cards with the highest ATK/DEF/Level more individualist? GitHub / Yu-Gi-Oh-Analytics-cpes-project1, May 2025.

[2] Farès Blibeche. Yu-Gi-Oh!: LLM-annotated card interaction dataset. GitHub / Yu-Gi-Oh-LLM-Annotation, May 2026.

[3] Farès Blibeche. Yu-Gi-Oh! complete dataset. GitHub / Yu-Gi-Oh-Scrapping, February 2026.

[4] YGOProDeck. YGOProDeck API. ygoprodeck.com / api-guide, 2026.

Appendix A. Variable Codebook

Table A1: Description of all variables used in the analysis.

Source	Variable	Description	Type
Cards API	`type`	Monster type (Dragon, Warrior, …)	Categorical
	`attribute`	Attribute (DARK, LIGHT, FIRE, …)	Categorical
	`frametype`	Frametype (effect, fusion, …)	Categorical
	`level`	Monster level (0–12)	Integer
	`atk`	ATK value (0–5000)	Integer
	`defe`	DEF value (0–5000)	Integer
Shared Features	`sharedType`	1 if both cards share monster type	Binary
	`sharedAttribute`	1 if both cards share attribute	Binary
	`sharedFrametype`	1 if both cards share frametype	Binary
Stat diff.	`lvDiff`	$\lvert \text{level}_A - \text{level}_B \rvert$, standardised	Continuous
	`atkDiff`	$\lvert \text{ATK}_A - \text{ATK}_B \rvert$, standardised	Continuous
	`defDiff`	$\lvert \text{DEF}_A - \text{DEF}_B \rvert$, standardised	Continuous
Annotation	`interactCount`	Mean total interactions per pair	Continuous
	`repeatCount`	Number of annotations of pair, standardised	Continuous
	`repeated`	1 if pair was annotated more than once	Binary
	`stability`	1 if all pair annotations agree	Binary

Appendix B. Annotation Quality Diagnostics

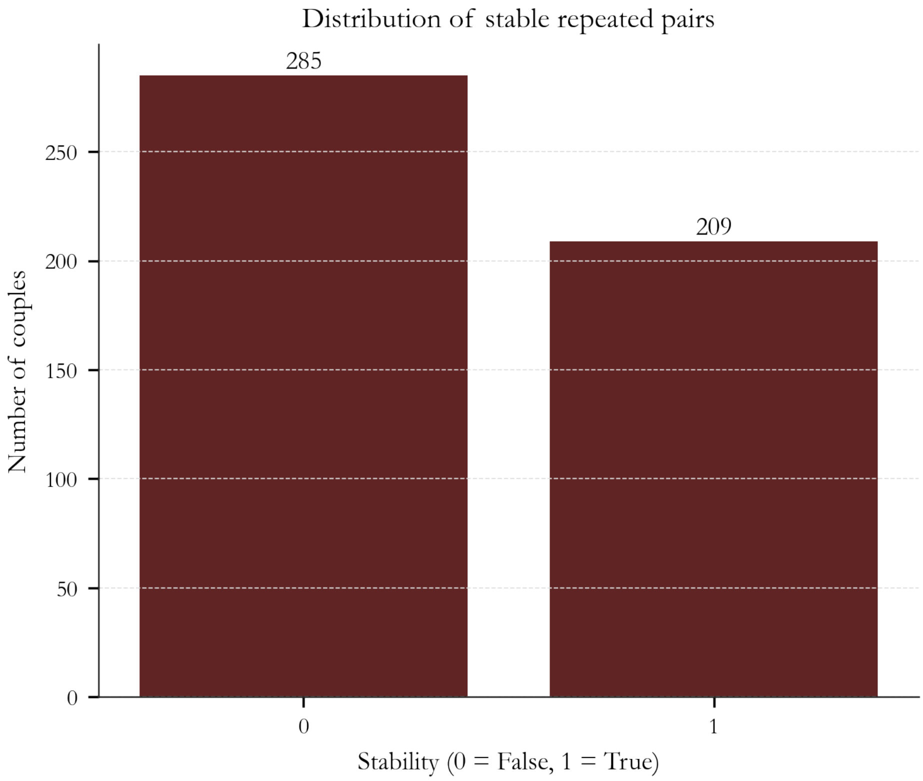


Fig. A1: Distribution of *stability* among repeated pairs (0: inconsistent, 1: consistent).

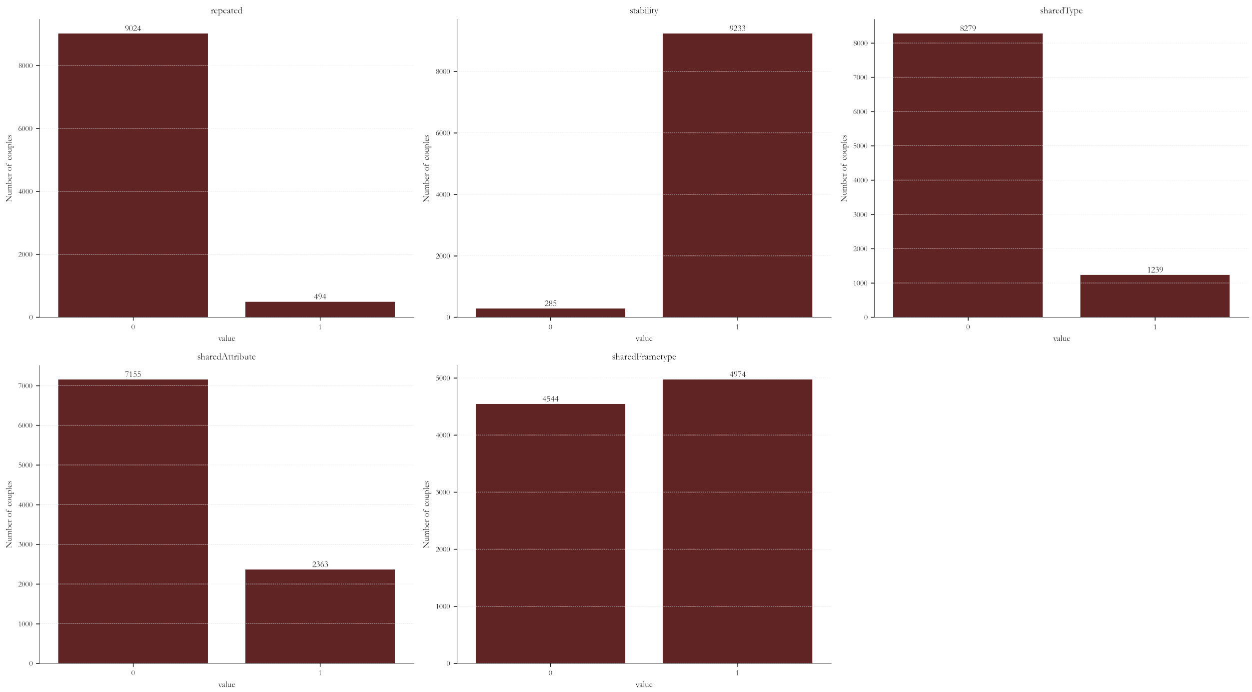


Fig. A2: Distributions of binary features (*SameType*, *SameAttr*, *SameFrame*) in the monster-pair subset used for regression.

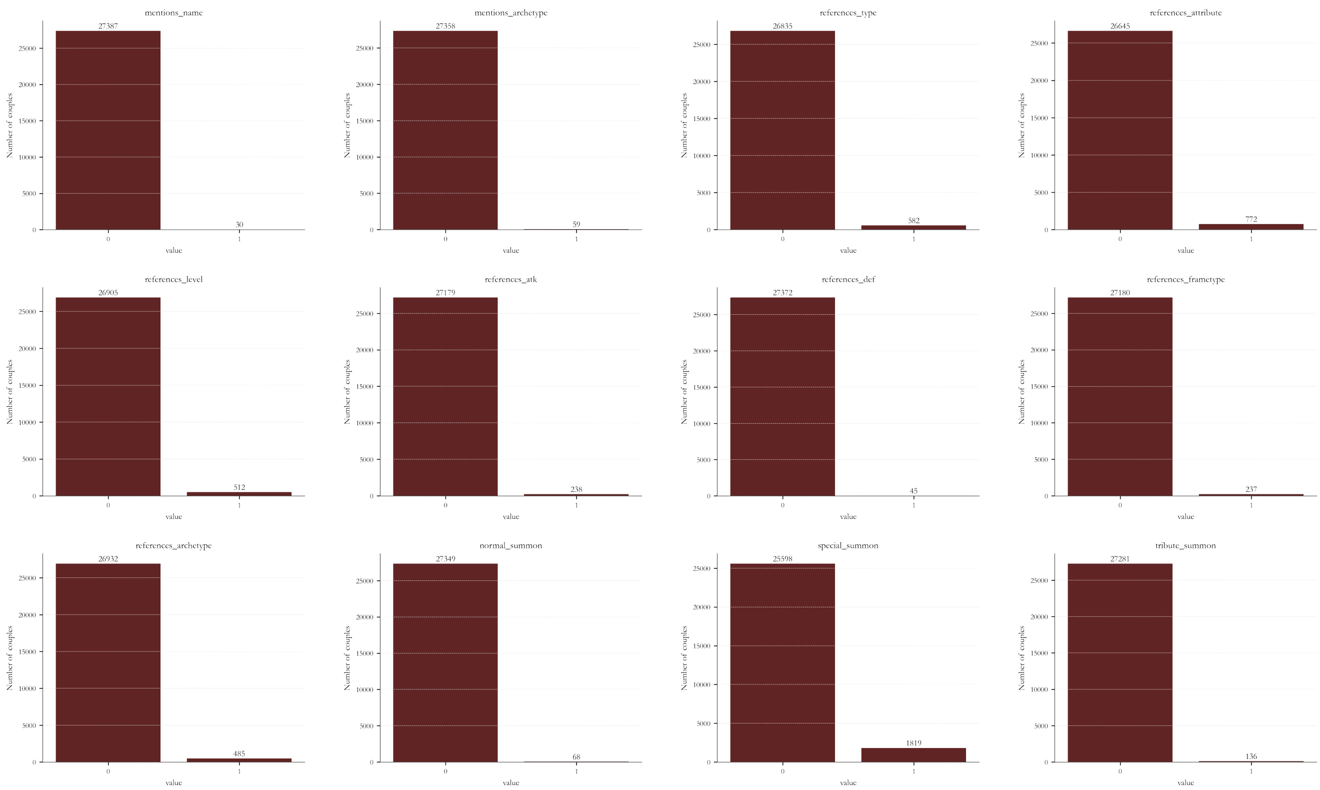
Appendix C. LLM Annotation Prompt Summary

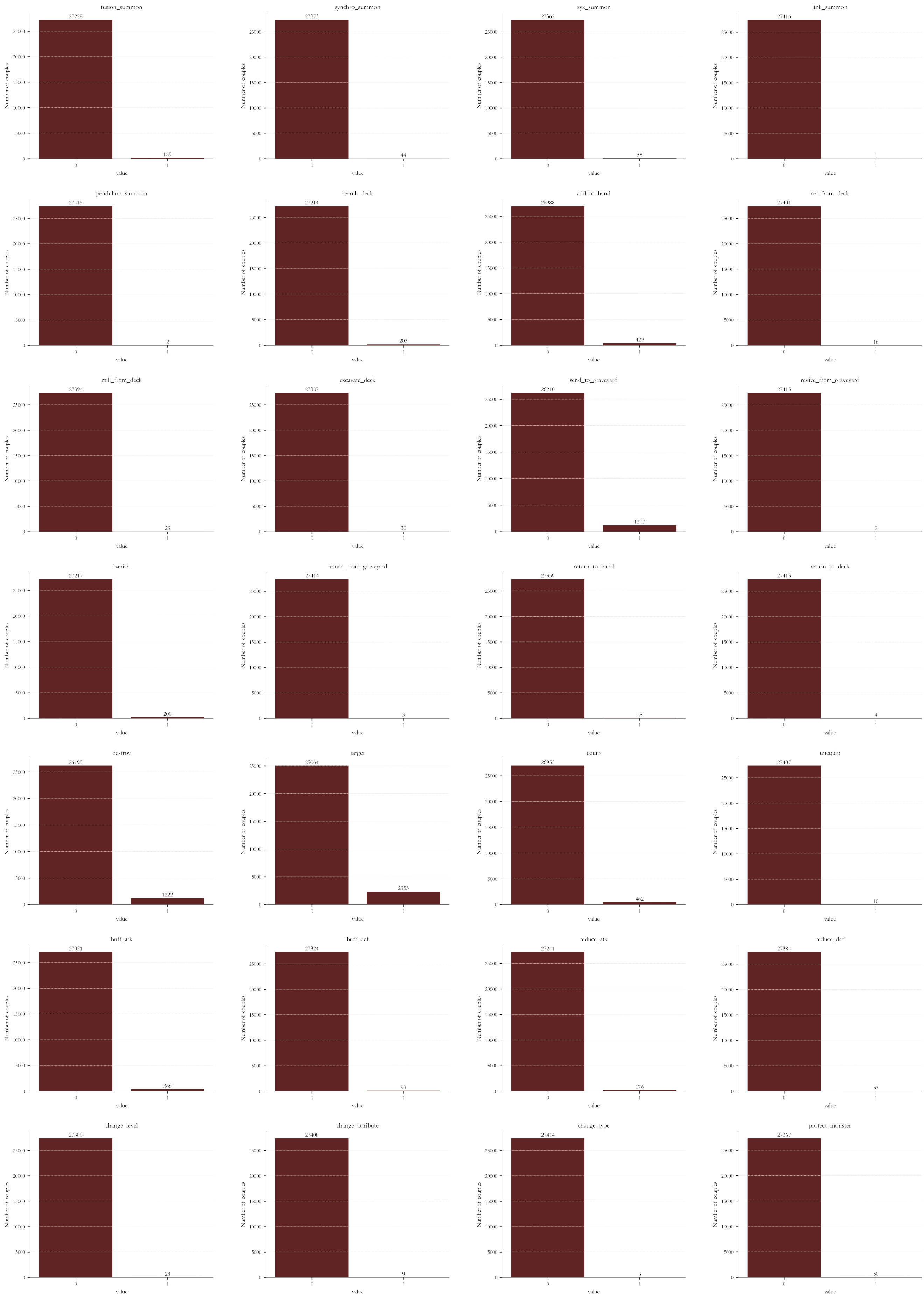
The LLM (Llama 3.1 8B) was prompted to identify directed interactions from Card A to Card B only, under the assumption that both cards are used from the same player. The model was provided with:

- The full effect text and data (type, attribute, frametype, level, ATK, DEF) of both cards.
- A glossary of 56 allowed relation types (e.g. `special_summon`, `destroy`, `references_type`, `buff_atk`, `negate_effect`).
- 13 annotated examples covering edge cases such as opponent-only effects and generic mass effects.

The model returned a JSON object listing applicable relation types with supporting evidence text and a confidence score. The full procedure is documented in `automated_annotator.ipynb`.

Appendix D. Per-Variable Interaction Distributions




fusion_summon
27228
189
synchro_summon
27373
44
xyz_summon
27362
55
link_summon
27416
1
pendulum_summon
27415
2
search_deck
27214
203
add_to_hand
26988
429
set_from_deck
27401
16
mill_from_deck
27394
23
excavate_deck
27387
30
send_to_graveyard
26210
1207
revive_from_graveyard
27415
2
banish
27217
200
return_from_graveyard
27414
3
return_to_hand
27359
58
return_to_deck
27413
4
destroy
26195
1222
target
25064
2353
equip
26955
462
unequip
27407
10
buff_atk
27051
366
buff_def
27324
93
reduce_atk
27241
176
reduce_def
27384
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3
protect_monster
27367
50
Number of couples
value

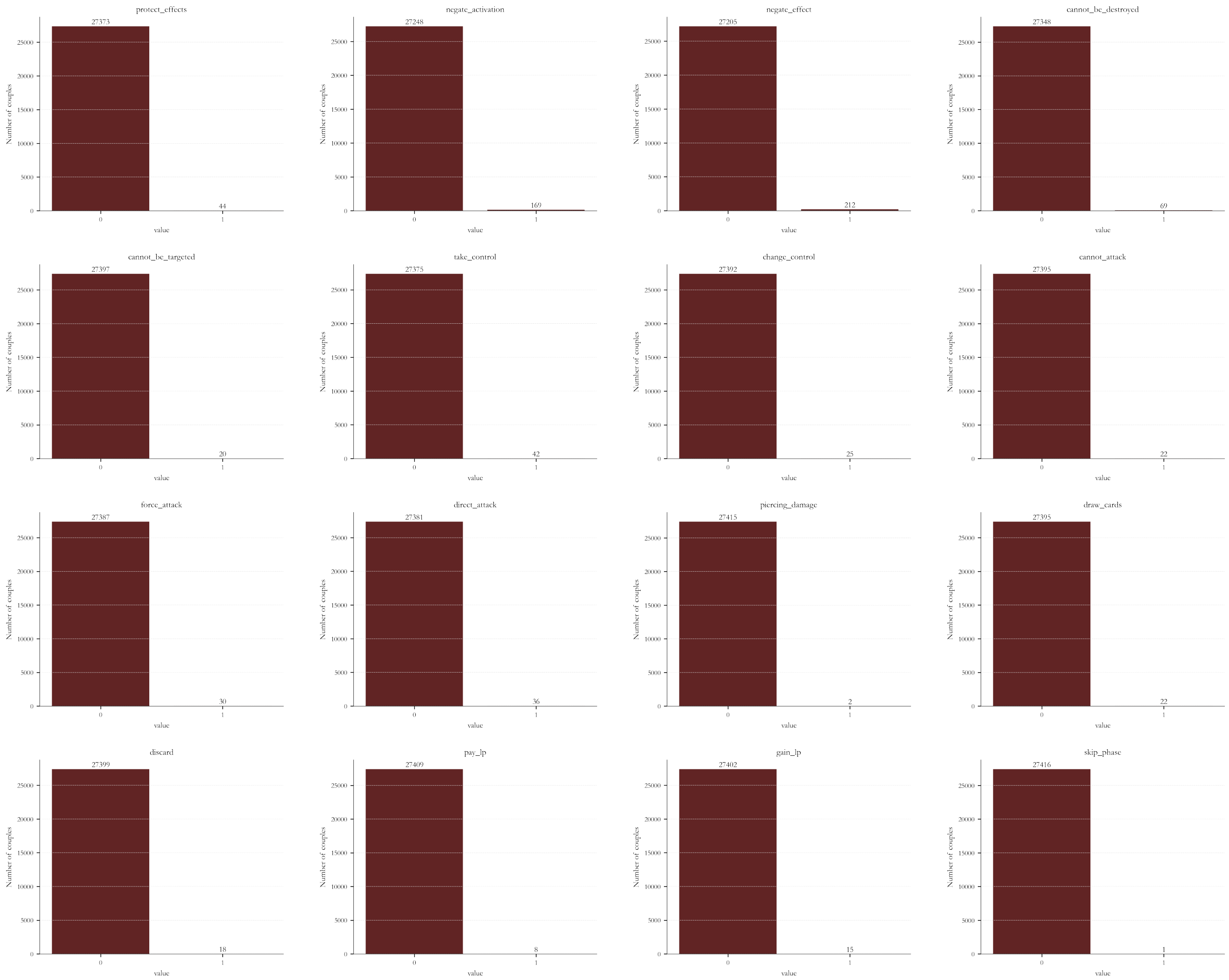

Fig. A3: Distribution of each of the 56 interaction types across the 27,417 annotated pairs. For each subplot, the left bar counts $x = 0$ (interaction absent), the right bar counts $x = 1$ (interaction present).